

1.	(a) The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
		1	
	(b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$ $H = 168 \text{ (J } ^\circ\text{C}^{-1}) \approx 170 \text{ (J } ^\circ\text{C}^{-1})$	1M	
		1A	
		2	
	(c) (i) Infra-red (radiation) / thermal radiation	1A	
		1	
		1A	
		1	
	(ii) Remote control / infra-red (IR) thermometer / night vision camera		
		1	

2. (a) $s = ut + \frac{1}{2}at^2$
 $0 = 5t + \frac{1}{2}(-9.81)t^2$
 $t = 1.0194 \text{ s} \approx 1.02 \text{ s}$ or 0 s (rejected)

(b) speed of the bead at its maximum height = 2 m s^{-1}

$$\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$$

$$= 0.02 \text{ J}$$

(c) The bead and the trolley have the same horizontal velocity and thus travel the same horizontal distance (within the same time interval t).

Or
 $v = u + at,$
 $0 = 5 + (-9.81)(t/2)$
 $t = 1.02 \text{ s}$

1M

1A

2

Accept: $t = 1.00 \sim 1.02 \text{ s}$

1M

1A

2

1A

1A

2

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s.

(b) Displacement = total area of the graph

$$= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$$

$$= 25 \text{ m}$$

Position of the object is 25 m from point O on the right side.

(c) Net force acting on the object

$$= ma$$

$$= 0.4 \times \frac{15 - (-10)}{10 - 0}$$

$$= 1 \text{ N (towards right)}$$

1A

2

1M

1A

2

1M

1A

2

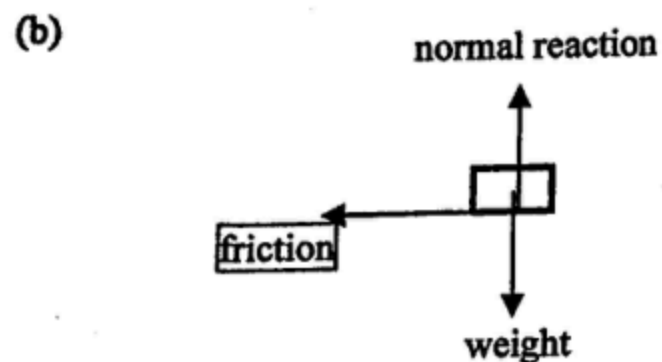
Or

$$v^2 = u^2 + 2as$$

$$15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$$

$$s = 25 \text{ m}$$

4. (a) $\omega = \frac{2\pi}{60}$
 $= \frac{\pi}{30}$ or $0.105 \text{ (rad s}^{-1}\text{)}$



- (c) The required centripetal force $F_C = m\omega^2 r$, where r is the distance of plasticine Q from the clock's centre. As m and ω are constant, smaller r gives smaller centripetal force F_C .
 The claim is correct.

1A

1

1A

Accept: adhesive force

1

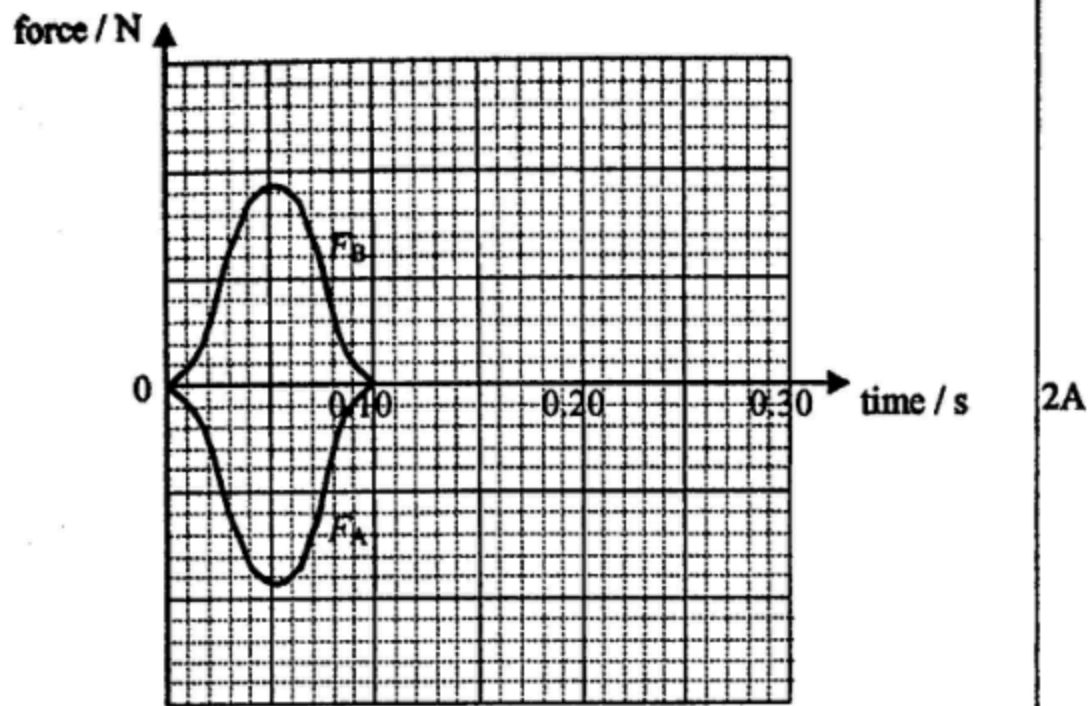
1A+1A

2

What are the components of the force acting on trolley A towards the right taken as positive), give trolley A a push and let it collide head-on with trolley B.

1A

Force-time graphs:



2A

The force acting on (the less massive) trolley B, i.e. F_B , is always equal (in magnitude and opposite in direction) to that acting on trolley A, i.e. F_A , as they are an action and reaction pair / according to Newton's 3rd law.

1A

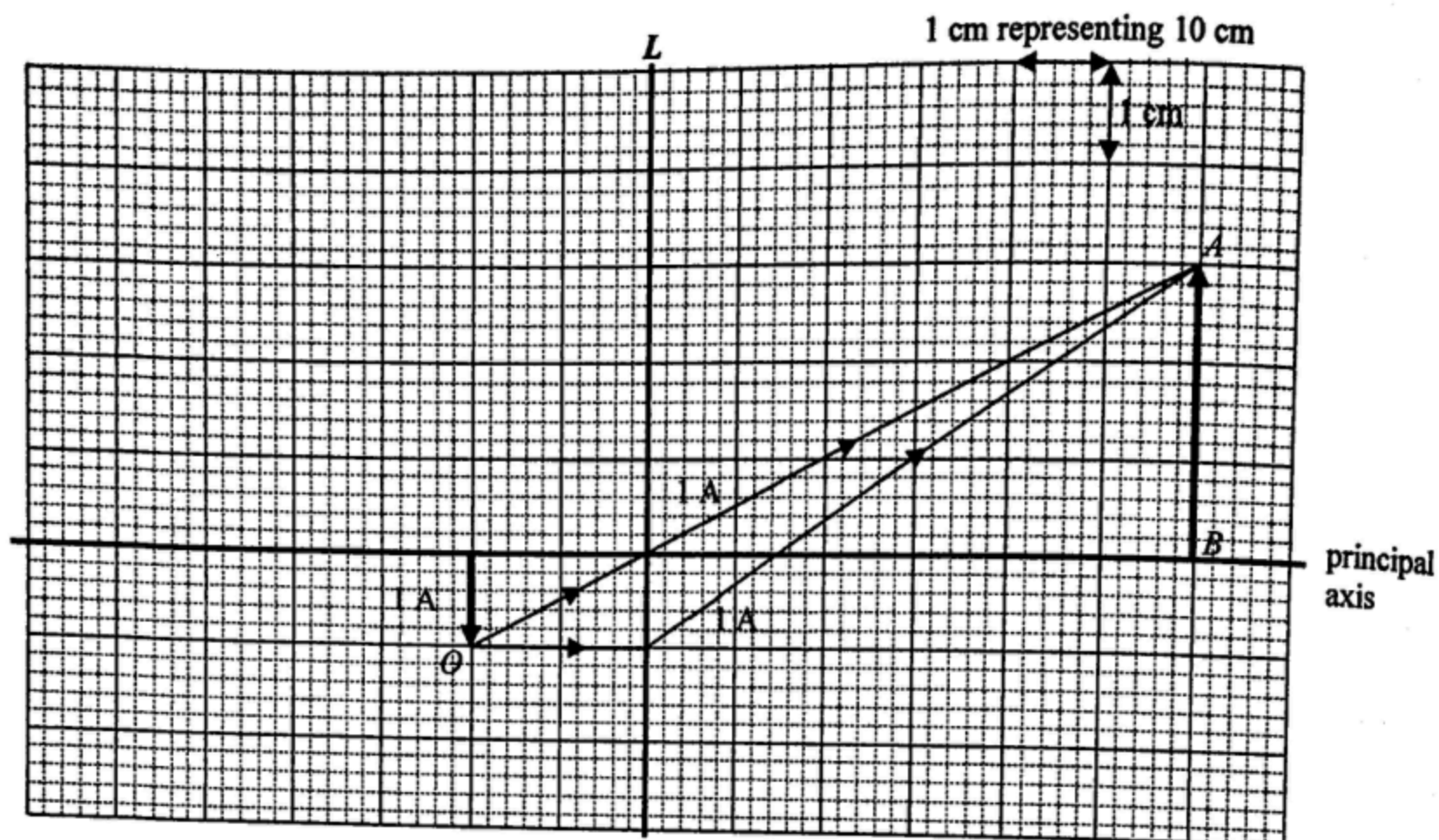
6. (a) Convex lens
because only a convex lens can form a real image.

1A
1A

Or
the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.

2

- (b) (i)(ii)



- (iii) Ray diagram method:
 $f = 15 \text{ cm}$

3

1A

Accept: 14 cm ~ 16 cm

Or

By $\frac{v}{u} = 3$,

$\frac{0.6}{u} = 3$, i.e. $u = 0.2 \text{ m}$

By $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$,

$\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$

$f = 0.15 \text{ m}$ or 15 cm

1

1A

1

1A

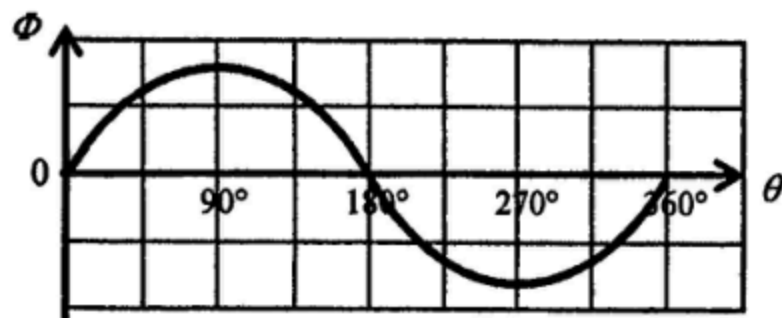
1

- (c) Move / adjust the support / smartphone closer to lens L.

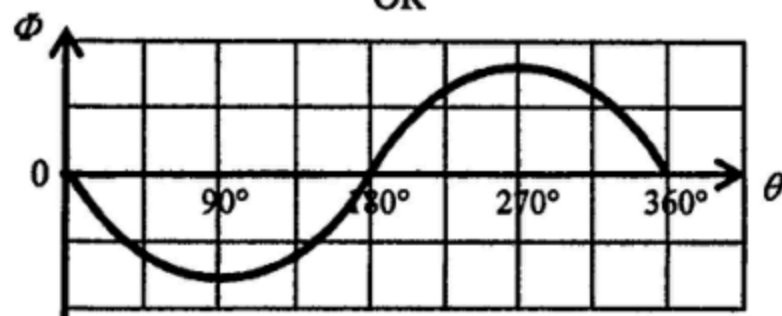
- (d) The box should be painted black inside.

7. (a) To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat.	1A	
	1	
(b) $R = \frac{V}{I} = \frac{3.5}{0.7}$ $= 5.0 \Omega (> 4.7 \Omega)$ The extra resistance (0.3Ω) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply.	1M 1A 1A	
	3	
(c) $I = 0.25 \text{ A}$ at $l = 30 \text{ cm}$ $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$ $= 9.0 \Omega$	1M 1A	Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$ e.c.f. from (b) Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$
	2	

8. (a)



OR



(b) (i) $\phi = BA \cos 0^\circ = 2.70 \times 10^{-4}$
 $B \times (9.00 \times 10^{-4}) \times 1 = 2.70 \times 10^{-4}$
 $B = 0.300 \text{ (T)}$

(ii) $\varepsilon = \left| -N \frac{\Delta \phi}{\Delta t} \right|$
 $= \left| -100 \times \frac{2.70 \times 10^{-4}}{0.01} \right|$
 $= 2.70 \text{ V}$

2A

2

1M

1A

2

1M

1A

2

Accept: -2.70 V

$$pV = nRT$$

$$1.01 \times 10^5 \times V = 1 \times 8.31 \times 273$$

$$V = 2.2462 \times 10^{-2} \text{ m}^3 \approx 2.25 \times 10^{-2} \text{ m}^3$$

(ii) Number of molecules in 1 m^3

$$= \frac{N_A}{\text{volume of one mole of molecule}}$$

$$= \frac{6.02 \times 10^{23}}{2.246 \times 10^{-2}}$$

$$= 2.680120 \times 10^{25} \approx 2.68 \times 10^{25}$$

(b)

$$pV = \left(\frac{N}{N_A} \right) RT$$

$$2.75 \times 10^{-10} \times 1 = \left(\frac{1.00 \times 10^{11}}{6.02 \times 10^{23}} \right) \times 8.31 \times T$$

$$T = 199.2178 \text{ (K)} \approx 199 \text{ (K)}$$

1M

1A

Accept: $0.0220 \sim 0.0225 \text{ m}^3$

2

1M

1A

e.c.f. from (a)(i)

Or

$$pV = nRT$$

$$1.01 \times 10^5 \times 1 = n \times 8.31 \times 273$$

$$n = 44.5$$

$$N = 44.5 \times 6.02 \times 10^{23}$$

$$= 2.68 \times 10^{25}$$

2

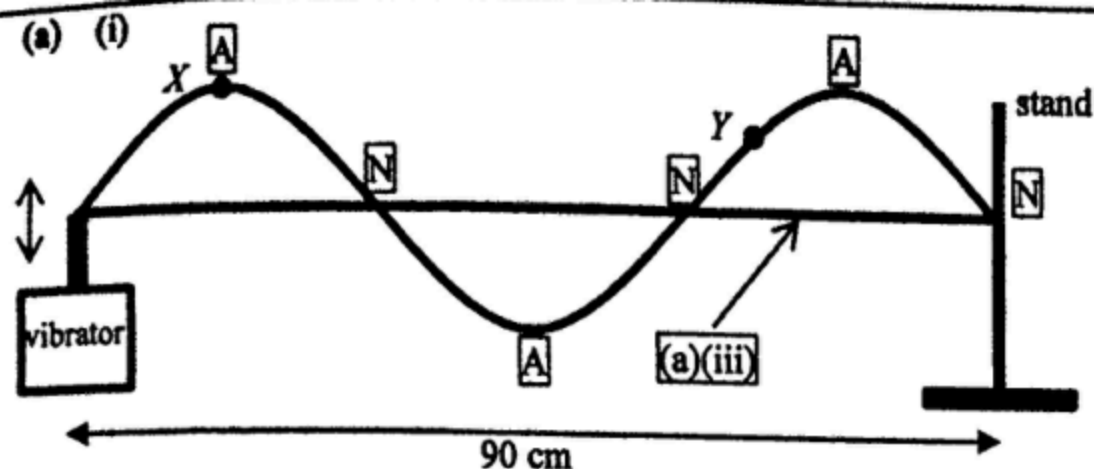
1M

1A

Accept: $199 \sim 200 \text{ K}$

2

10. (a) (i)



- (ii) Particle X vibrates with a larger amplitude than particle Y while they are vibrating in phase.

- (iii) Correct sketch of a straight string

- (iv) Original wavelength

$$\lambda_1 = \frac{0.90}{1.5} = 0.60 \text{ m at } f_1 = 100 \text{ Hz}$$

For the next stationary wave pattern,

$$\lambda_2 = \frac{0.90}{2} = 0.45 \text{ m}$$

By $v = f\lambda$,

$$100 \times 0.60 = f_2 \times 0.45$$

$$f_2 = 133.33 \text{ Hz} \approx 130 \text{ Hz}$$

(b) (i)



- (ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms.

- (iii) Any TWO:
stationary vs travelling
transverse vs longitudinal
energy confined in a string vs energy being transferred through air

2A

2

1A

1A

2

1A

1

1M

1A

2

1A

1

1A

1A

2

2A

2

Or

$$f_2 = 100 \times \frac{4}{3}$$

$$= 133.33 \text{ Hz} \approx 130 \text{ Hz}$$

$$= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$$

$$(ii) \quad E_{\text{out}} = (680 \times 10^6) \times (24 \times 3600) \\ = 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$$

$$(iii) \quad \text{Efficiency} = \frac{E_{\text{out}}}{E_{\text{in}}} \times 100\% \\ = \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\% \\ = 40.8\% \approx 41\%$$

$$(b) \quad (i) \quad P = VI \\ 680 \text{ MW} = 400 \text{ kV} \times I \\ I = 1700 \text{ A} = 1.7 \text{ kA}$$

$$(ii) \quad \text{Power loss} = I^2 R \\ = (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3) \\ = 13410 \text{ W} \approx 13.4 \text{ (kW)}$$

(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.

1M
1A

2

1M

1A

2

e.c.f. from (a)(i)(ii)

Accept: 41% ~ 42%

1M

1A

2

1M

1A

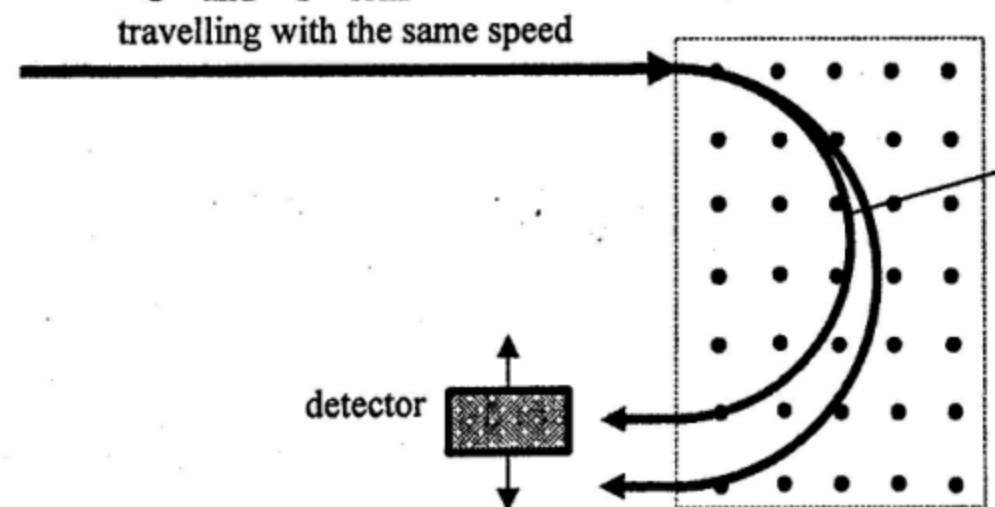
2

e.c.f. from (b)(i)

1A

1A

2

12. (a) (i) ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e}$ (or β) (ii) $k = \frac{\ln 2}{t_{\frac{1}{2}}}$ $= \frac{\ln 2}{5370}$ $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}\text{)} \approx 1.21 \times 10^{-4} \text{ (yr}^{-1}\text{)}$ (iii) $C = C_0 e^{-kt}$ $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$ $t = 6764.7 \approx 6800 \text{ (years)}$	1A 1 1M 1A 2 1M 1A 2	e.c.f. from (a)(ii) Accept: 6760 ~ 6820 (years) Or $(13.6)\left(\frac{1}{2}\right)^n = 6.0$ $n \log\left(\frac{1}{2}\right) = \log\left(\frac{6.0}{13.6}\right)$ $n = 1.181$ $t = n \times 5730 = 6764.7 \text{ (years)}$ $\approx 6800 \text{ (years)}$
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number. (ii) ${}^{12}\text{C}^{+1}$ and ${}^{14}\text{C}^{+1}$ ions travelling with the same speed  detector path of ${}^{12}\text{C}^{+1}$ ions (iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.	1A 1 1A 1A 1A 2	Or same number of protons but different number of neutrons